UF AI DAYS · PLENARY TALK

Tue., Oct. 27, 2026 · 2:40–2:55 PM · Reitz Union Grand Ballroom

AI Research, Teaching, and Learning Day

Restoring the Milky Way

Scaling AI-assisted light pollution planning across North Central Florida

- The Dark Sky Simulator: satellite AI, geospatial mapping and scenario modeling
- Test lighting policy before it is written into law

Jay Rosen
Research Software Engineer III
University of Florida College of Education · jayrosen@ufl.edu

Geospatial AI · Computer Vision · Remote Sensing · Decision Support · Policy Simulation



Photo: Milky Way and urban skyglow over Paynes Prairie, 2022. Jay Rosen

THE PROBLEM

Skyglow travels: one city's lights wash out a whole region



Gainesville's skyglow and Comet NEOWISE from Paynes Prairie, July 2020. Photo: Jay Rosen

- Light scattered by air and haze brightens the sky **tens of km** away
- Harms wildlife, sleep and health, and wastes energy



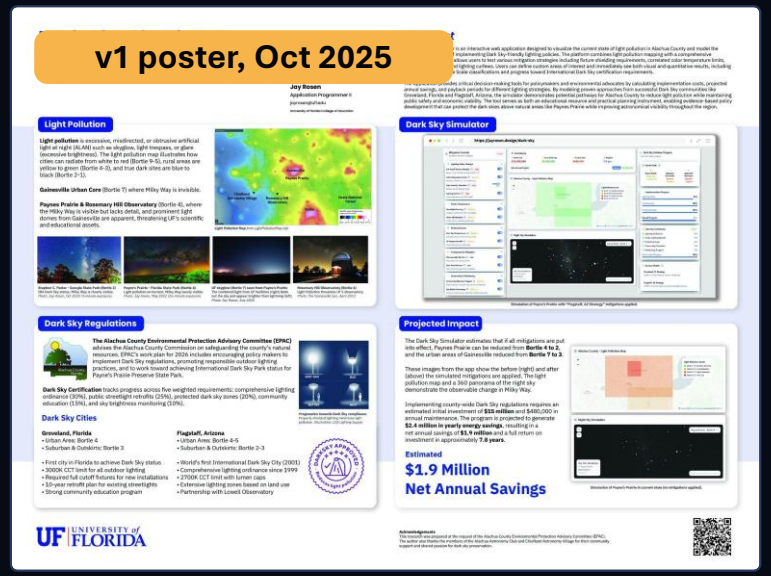
International Dark Sky Park, October 2020. Photo: Jay Rosen

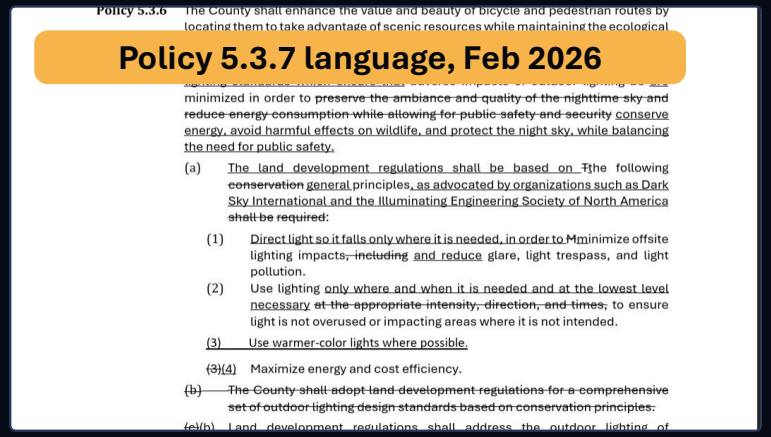
- Threatens UF's dark sites: **Rosemary Hill Observatory** and **Chiefland Astronomy Village**
- **Preventable**: shield, aim, warmer color, less brightness, dim late at night

BACKSTORY

From astrophotography to county policy

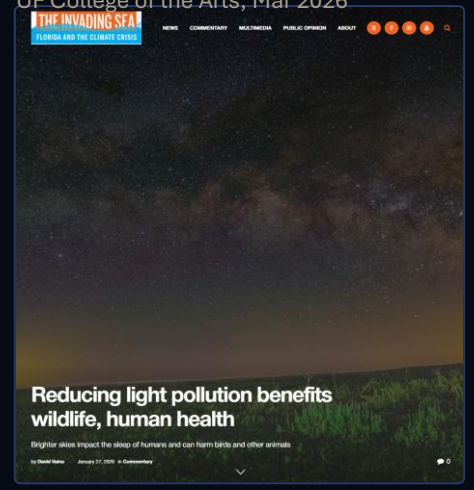
- **2020–25** Photographing the night sky under Gainesville's light dome
- **Sep 2025** Public comment at Alachua County's advisory committee (EPAC); v1 simulator built in a month
- **Jan 2026** EPAC votes unanimously to work toward dark-sky policy
- **Feb 2026** Policy 5.3.7 amendment (Comprehensive Plan, Conservation and Open Space Element) goes to the County Commission
- **2026** Expanding into **Levy County**, home to Rosemary Hill and the Chiefland Astronomy Village
- **Next** UF Astronomy (Dr. Elizabeth Lada) asks where a future observatory should go







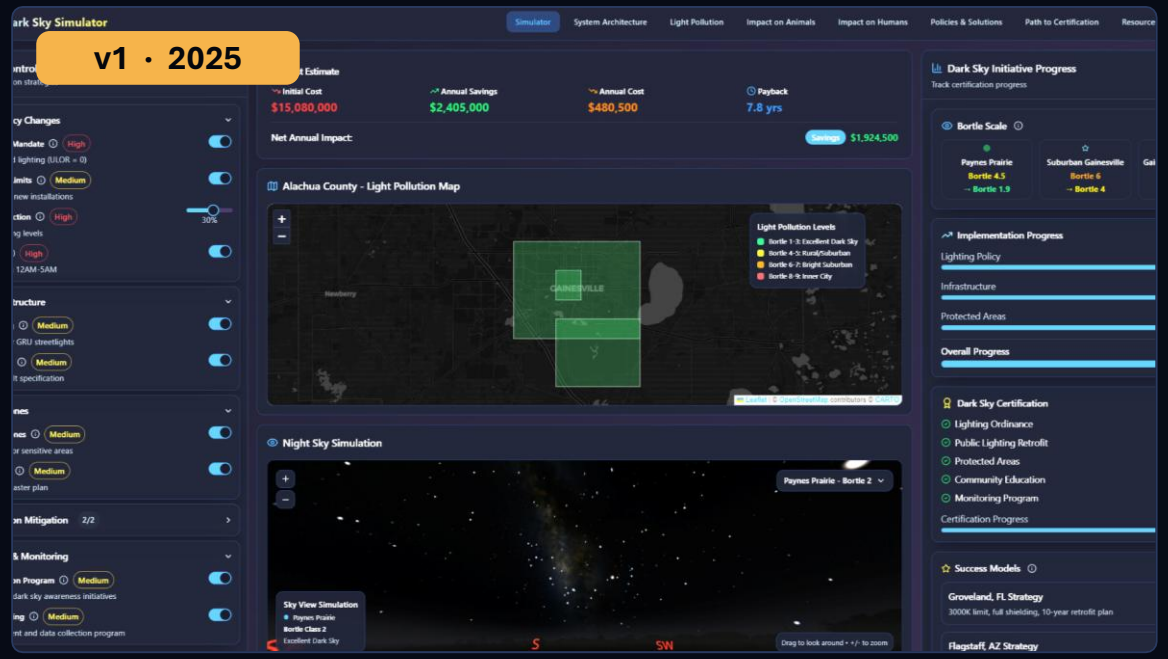
UF College of the Arts, Mar 2026

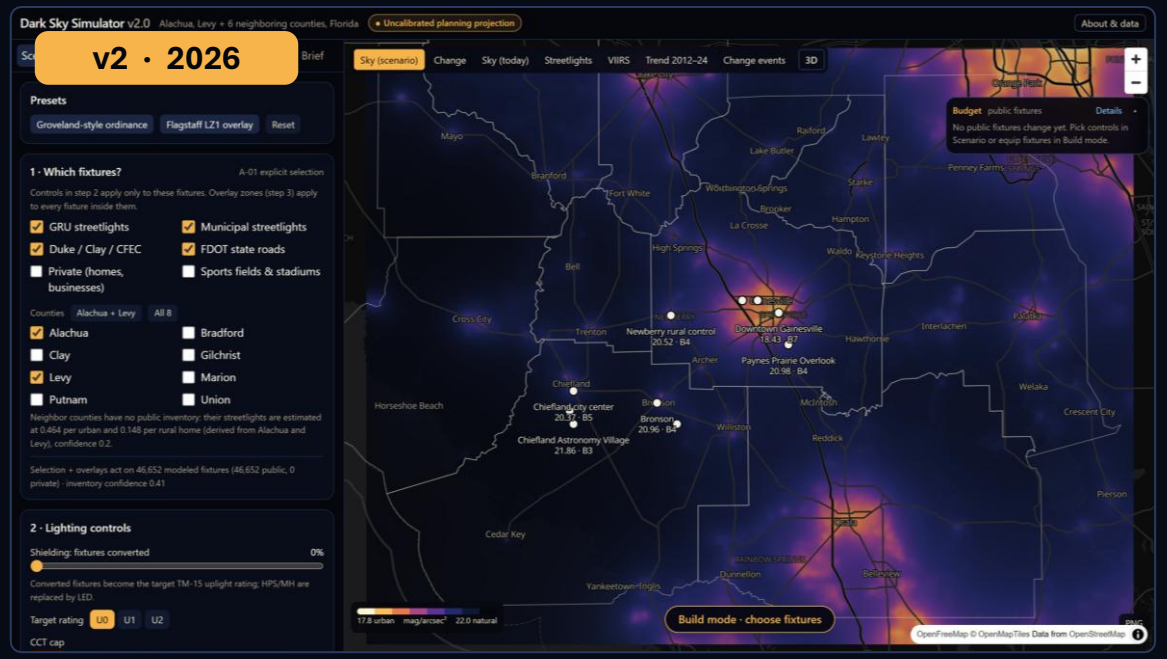


The Invading Sea, Jan 2026

WHY A VERSION 2

v1 told a story. v2 has to answer “How do you know?”



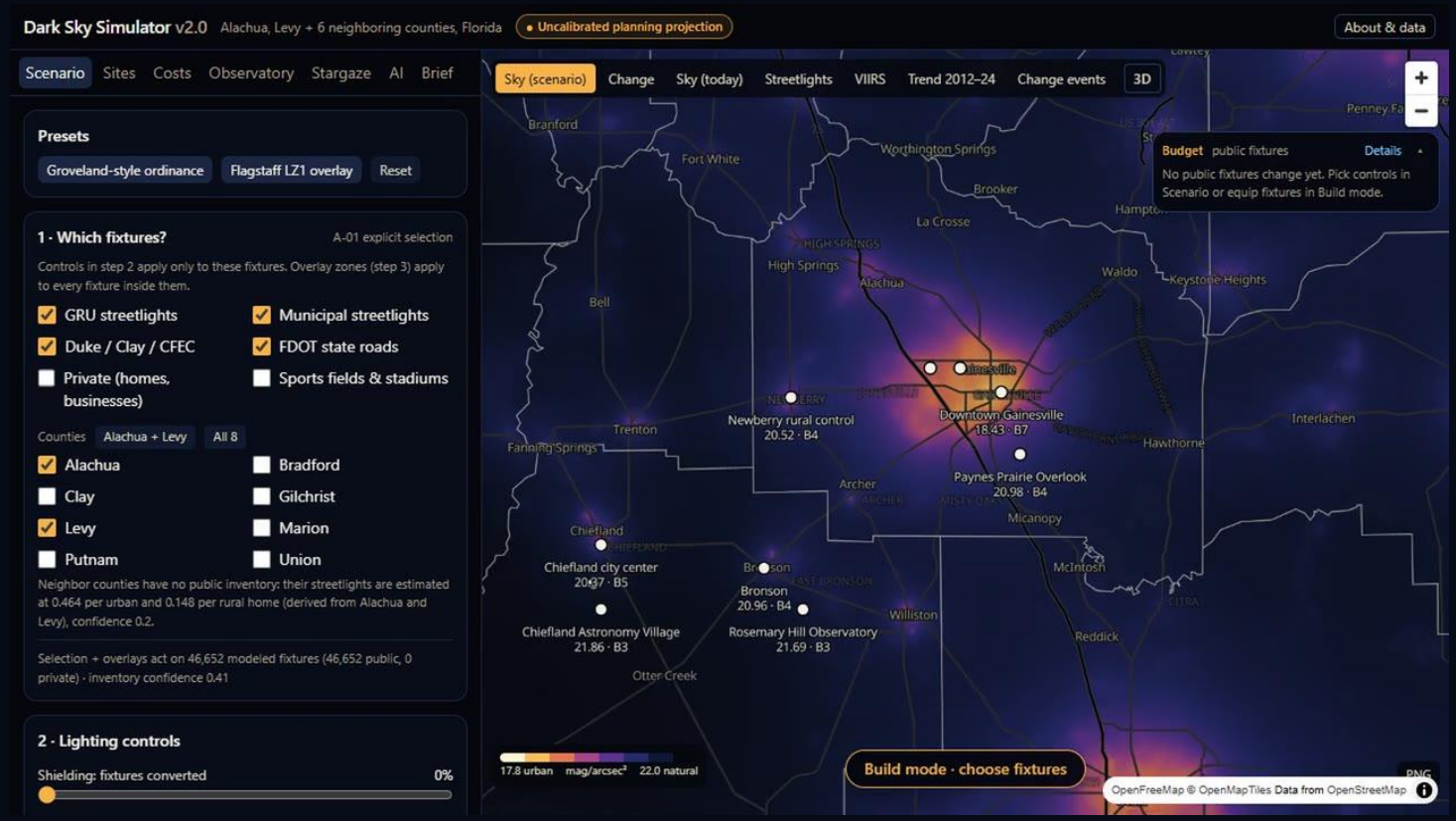


- Hand-tuned multipliers, built in a month
- Persuasive headline numbers (“$1.9M a year saved”) that nobody could audit

- Physics for the sky, public data for the lights, **AI only where data can teach**
- Every number labeled a planning projection; runs on a phone, no server

v2 at a glance

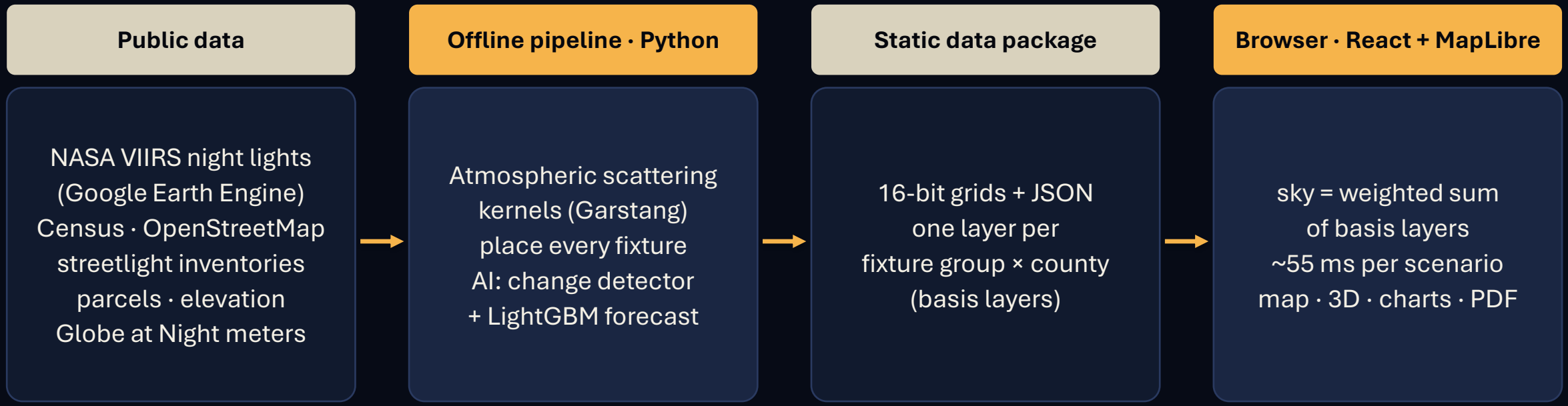
- **8 counties**: Alachua, Levy and six neighbors, ~0.4 km cells
- **~817,000** modeled light sources: streetlights, homes and businesses, sports fields
- Map: modeled zenith sky brightness (mag/arcsec^2); yellow = urban, black = near natural
- Seven tabs: Scenario · Sites · Costs · Observatory · Stargaze · **AI** · Brief



Scenario tab with today's modeled sky. Amber badge on every screen: “Uncalibrated planning projection”.

HOW IT WORKS

Heavy work offline, instant scenarios in the browser



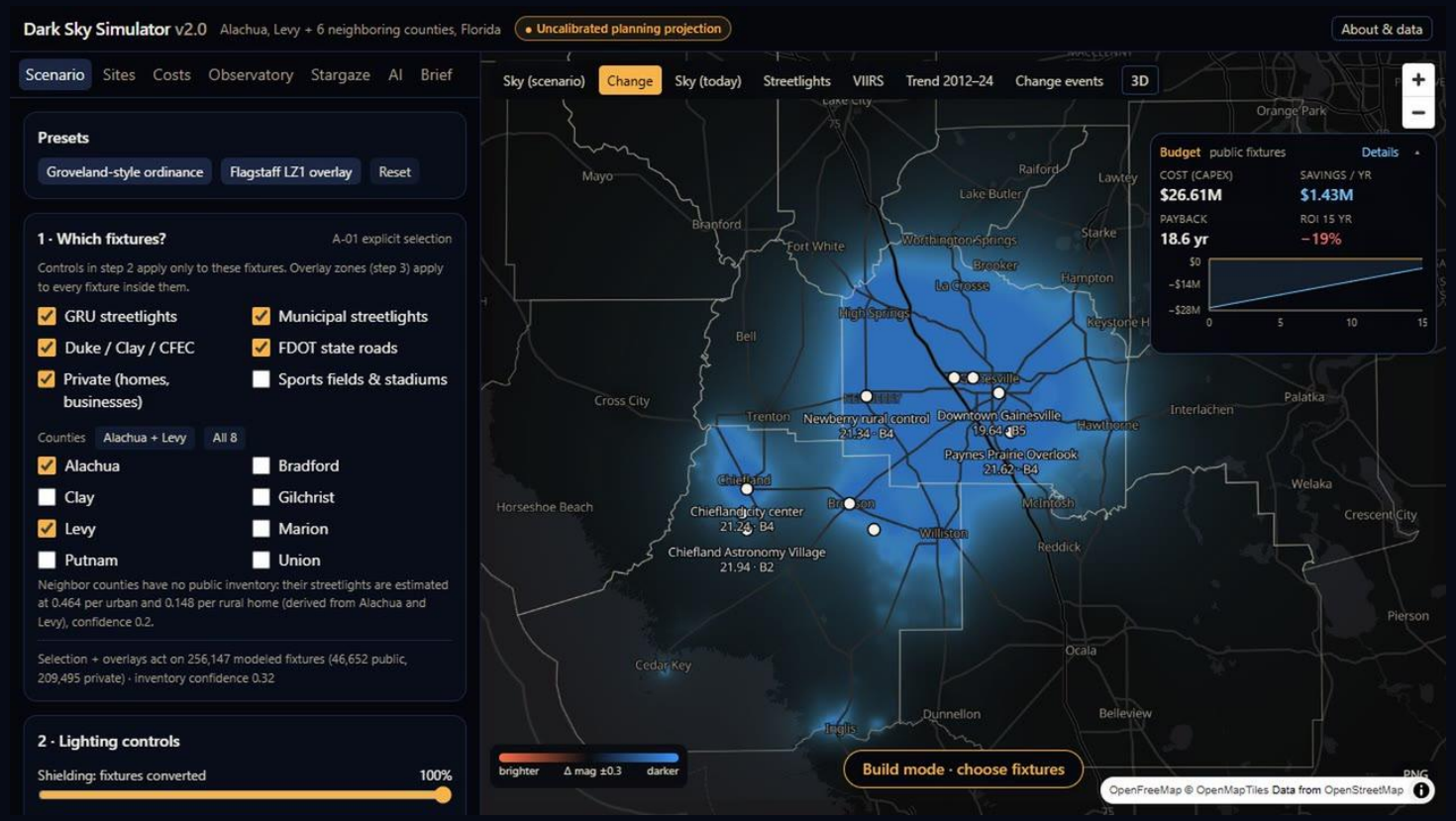
$$L^{\mathrm{scn}}(x) = \alpha \sum_{c,\,t} \Phi^{\mathrm{scn}}_{c,t}\, B_{c,t}(x), \qquad m = 12.6 - 2.5 \log_{10}\left(L_{\mathrm{art}} + L_{\mathrm{nat}}\right)$$

- Skyglow adds linearly, so any ordinance is a re-weighting of precomputed layers: a phone can explore it instantly
- The sky is a published single-scattering model (Garstang 1986), **not** a neural network

SCENARIO

One click: a Groveland-style ordinance

- Groveland: Florida's first dark-sky city
- Full shielding (U0), **3000 K** cap, dim to 50% from 22:00 to 06:30
- Paynes Prairie: **20.98 → 21.62** mag/arcsec^2 if every light complies; **21.30** with public lights alone
- Blue = darker sky, and it crosses county lines
- Public cost: **$26.6M**, $1.43M/yr saved, 18.6-yr payback



Change layer: where the sky gets darker (blue) under the ordinance; Budget card at top right.

SCENARIO · COSTS

What each government pays, and saves

Right-sized public streetlights: **fully shielded 3000 K LEDs at 30% lower output**, no smart controllers. Savings = energy + avoided maintenance, at each utility's rates.

Alachua County

all public streetlights (43,517)

$15.3M	$2.55M	6.0 yr	+$23.0M
one-time cost	saved / year	payback	after 15 years

University of Florida*

4,100 campus pole lights

$984K	$198K	5.0 yr	+$2.0M
one-time cost	saved / year	payback	after 15 years

City of Gainesville

GRU streetlights (30,000)

$7.20M	$1.45M	5.0 yr	+$14.6M
one-time cost	saved / year	payback	after 15 years

State roads (FDOT), Alachua

1,500 state-road lights

$900K	$113K	8.0 yr	+$795K
one-time cost	saved / year	payback	after 15 years

Levy County

all public streetlights (3,135)

$1.88M	$228K	8.3 yr	+$1.54M
one-time cost	saved / year	payback	after 15 years

City of Chiefland

285 city streetlights

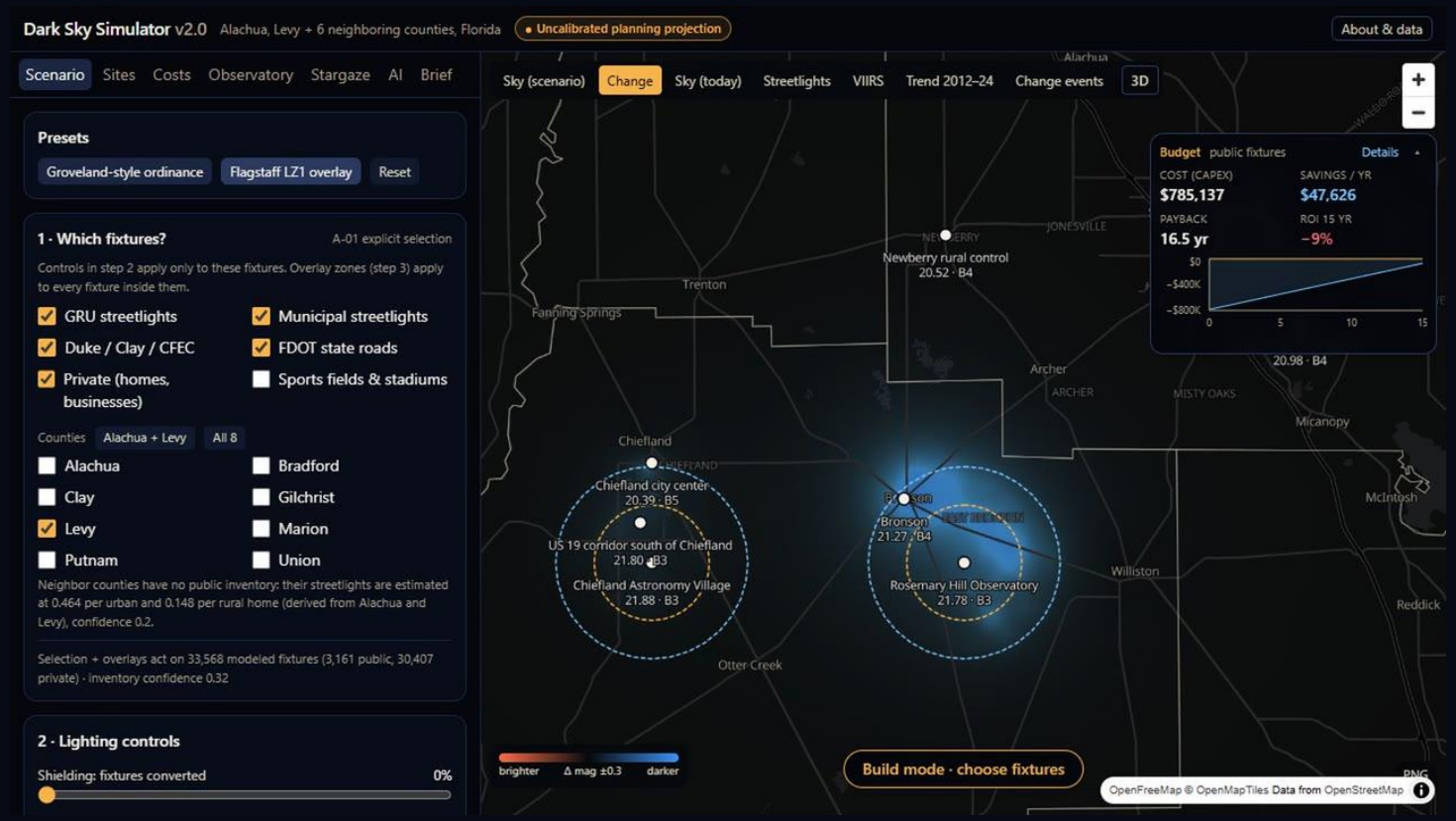
$171K	$21.5K	8.0 yr	+$152K
one-time cost	saved / year	payback	after 15 years

*UF: 4,100 campus pole lights, about 60% already LED (UF News, 2023), run through the simulator's cost engine at GRU rates. Gainesville and FDOT are within Alachua's total.

SCENARIO

Or protect just the observatories

- Overlay zones: stricter rules in rings around **Rosemary Hill** and **Chiefland Astronomy Village**
- LZ0 core: shielded, amber, motion-only
- LZ1 buffer: shielded, ≤ 2700 K
- Flagstaff-style preset: about **$0.79M** in public retrofits
- A trade-off a commission can actually vote on



Flagstaff LZ1 overlay preset (3-mile core, 5-mile buffer) on the Change layer.

SEE IT

Same moonless night, two skies



All-sky view looking south, August 12, 2026, 22:00, new Moon. Illustrative, drawn from the computed sky brightness; not a photograph.

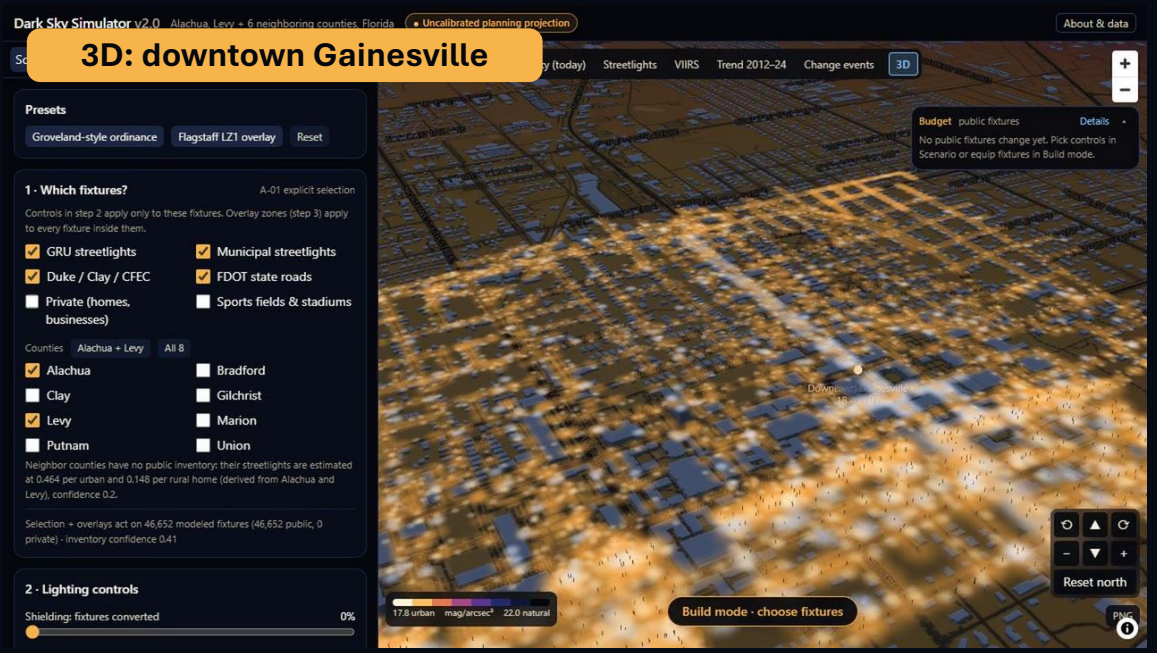
- **18.43** mag/arcsec^2 · Bortle 7
- Faintest naked-eye stars ≈ magnitude 4.3

- **21.69** mag/arcsec^2 · Bortle 3
- Faintest stars ≈ 6.5: the Milky Way is back

SCENARIO

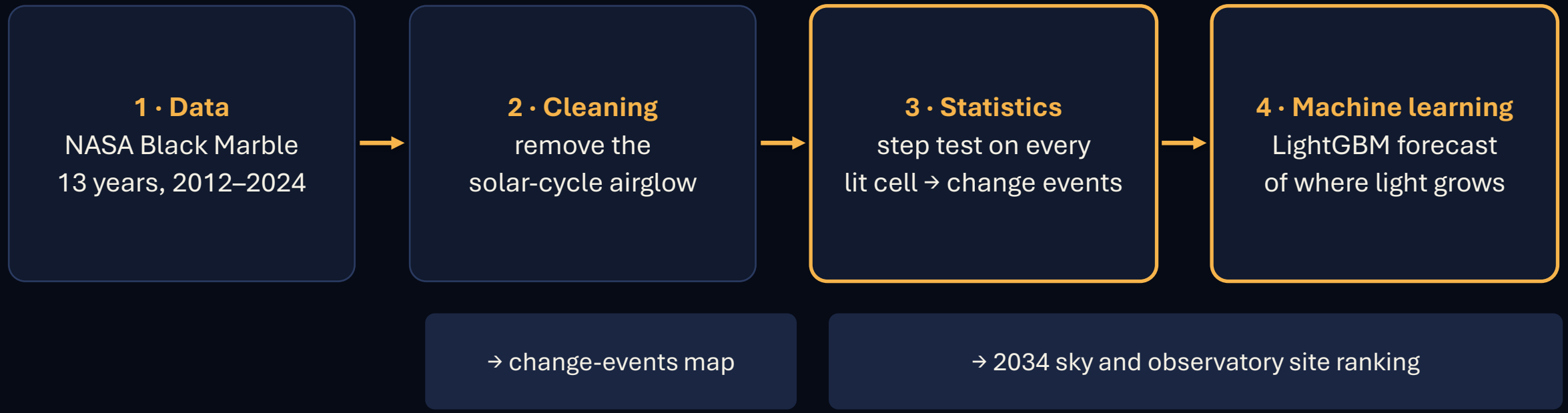
Build mode and 3D streets





- Catalog of **17 DarkSky-compliant fixture types**: spectrum, uplight rating, watts, installed cost
- Equip one: sky, budget, brief and install markers update
- Terrain, buildings and every modeled light at street level
- Mapped streetlights where they stand; modeled ones along the roads of their cell

Where AI is used, and where it isn't

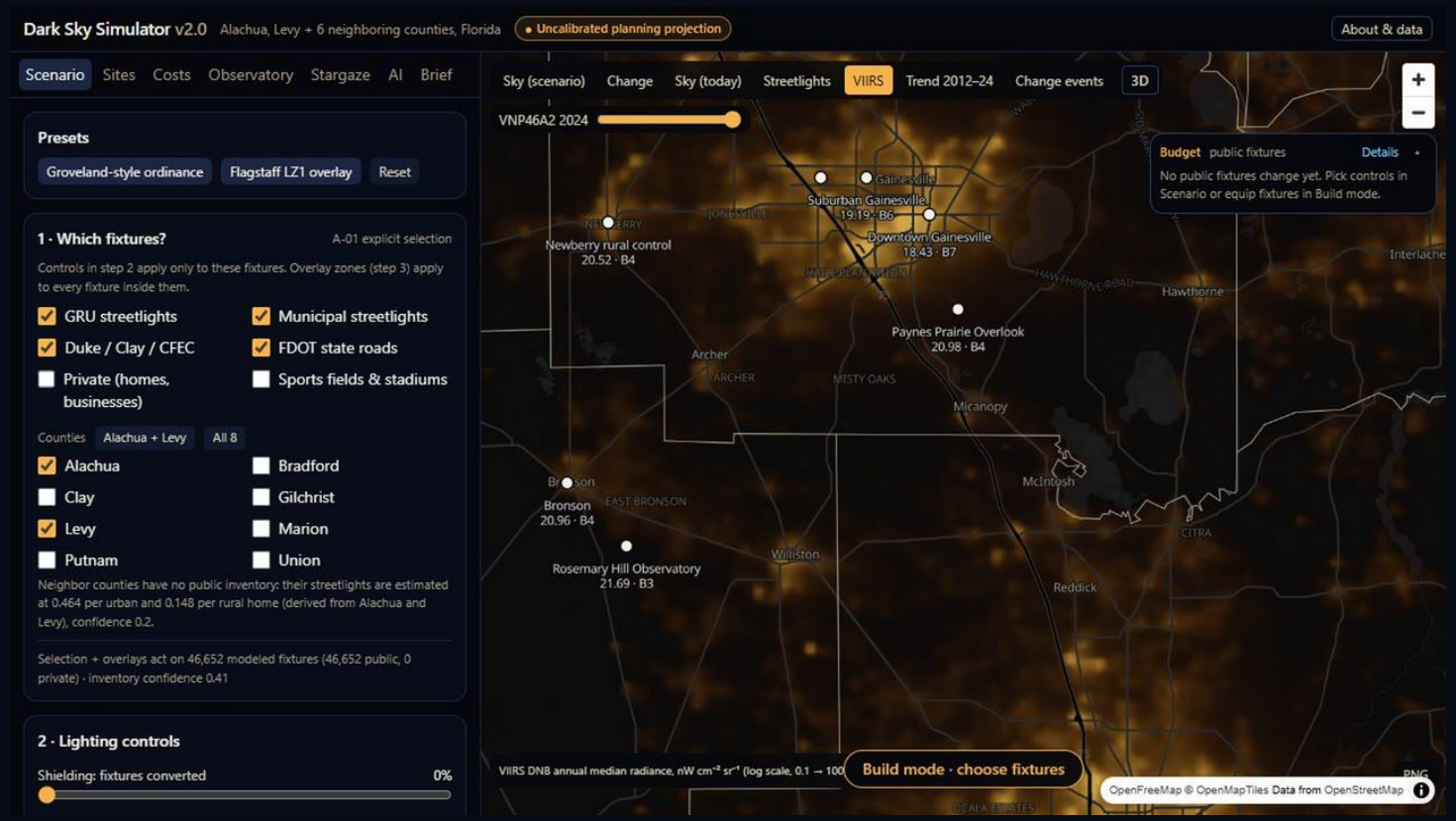


NOT AI: physics and explicit rules you can inspect

- **Sky brightness**: atmospheric scattering physics
- **Scenario effects**: published fixture rules
- **Site ranking**: weighted sum, weights you set
- **Costs**: unit prices × fixture counts

The data: 13 years of NASA night lights

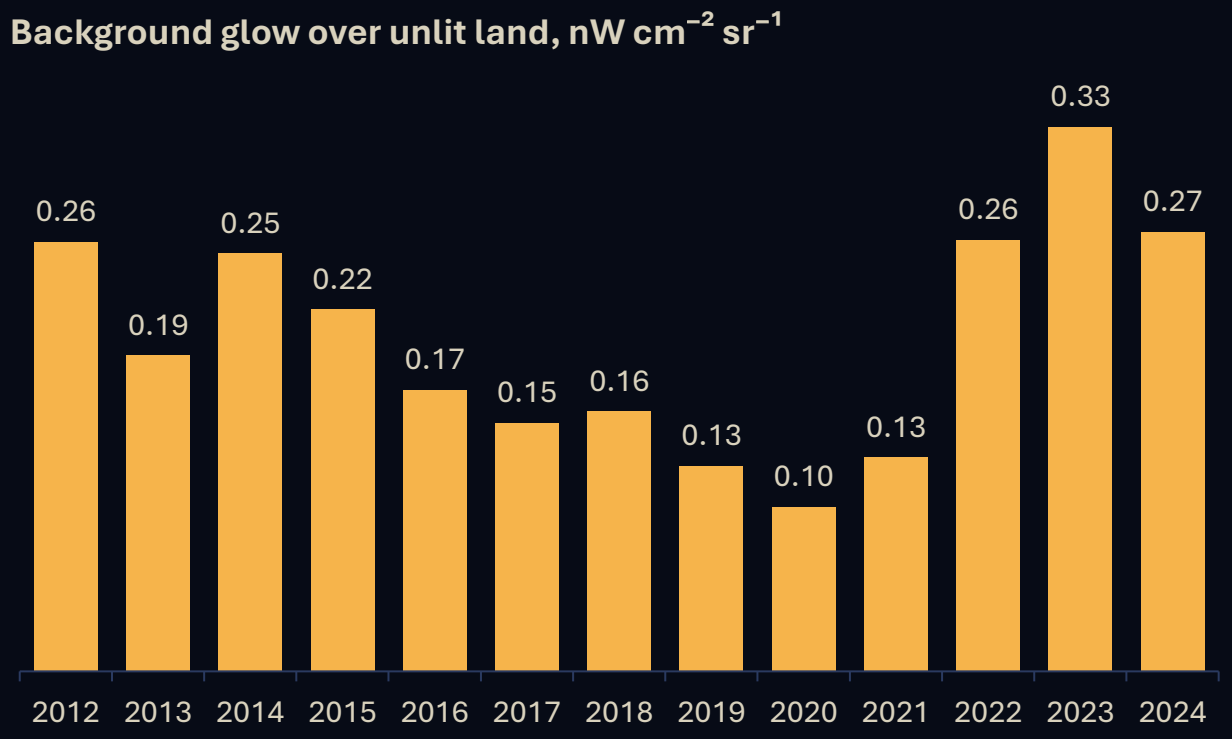
- NASA Black Marble (**VIIRS VNP46A2**): nightly, moonlight- and angle-corrected, ~500 m
- Keep cloud-free, high-quality nights; **one median per year** per cell, 2012–2024
- Pulled through **Google Earth Engine**'s free noncommercial tier in ~12 minutes
- The 2024 layer also tells the physics model where private lighting is



VIIRS layer: yearly median radiance with the airglow background removed; the slider steps through 2012–2024.

Lesson 1: check the data before the model

- First pull came back **10× too dark**: a scale factor applied twice
- Then forests and wetlands “brightened” **70–120%** from 2020 to 2023
- Cause: **airglow**, upper-atmosphere light that follows the 11-year solar cycle
- Fix: subtract each year's background, the median of the darkest fifth of the region, before any model sees the data



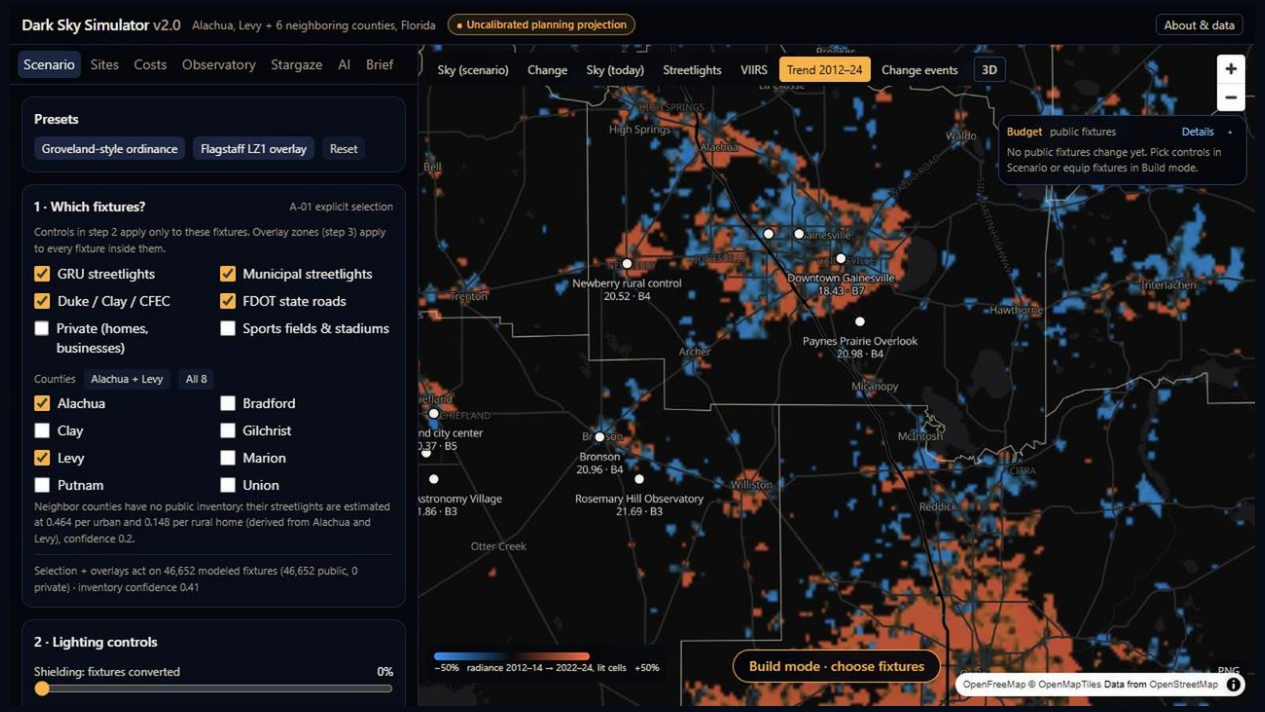
Solar minimum 2019–20: 0.10 · solar maximum 2023: 0.33

$$\tilde{L}_t(x) = \max\left(L_t(x) - b_t,\ 0\right), \qquad b_t = \operatorname{median}_{x \in \mathcal{D}} L_t(x)$$

The 19% estimate, checked with calibrated data

- AI analysis of NOAA night imagery, 2012 → 2024: **≈19% brighter** (v1)
- Re-measured in calibrated NOAA and NASA radiance: the **region around Alachua County rose ~20%**
- Inside the county line: **3–6%**; around Gainesville ≈5%. Growth concentrates along I-75 and around Ocala

Calibrated radiance, 2013 → 2024	NOAA EOG	NASA BM
Region around the county	**+20.5%**	**+12.8%**
Alachua County	+6.1%	+2.9%
Gainesville	+5.0%	+4.8%

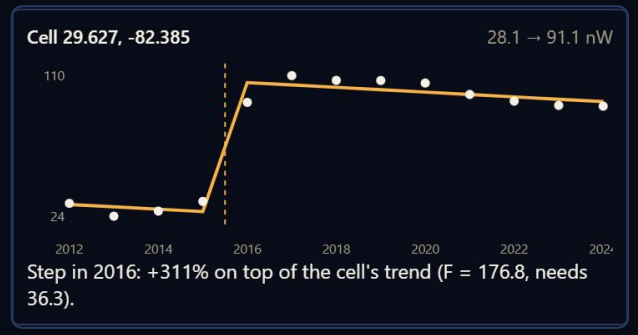


Trend 2012–24 layer: change from 2012–14 to 2022–24 for lit cells; growth along I-75 and around Ocala.

Finding where lighting changed: trend or step?

- For every lit cell: is a steady trend enough, or was there a **sudden step**?
- F-test of line vs line + step, at the best of 8 split years (2015–2022)
- Cutoff from simulating **200,000** noise-only cells: F ≥ 36.3, a false alarm **1 in 1,000**
- Region's shared movement removed first; steps must be ≥ 20%
- Our first detector flagged **2,000 of 2,000** simulated steady trends





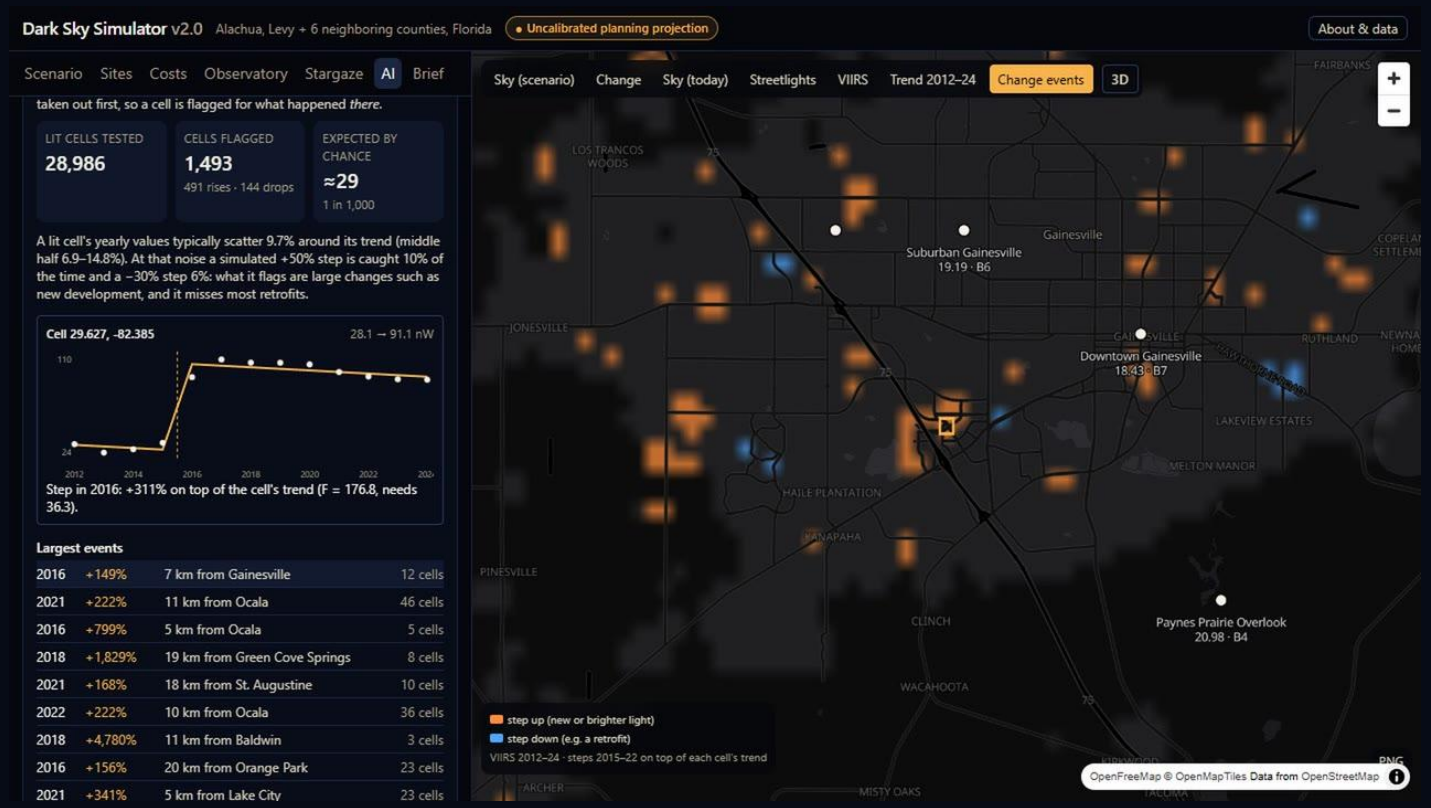
Steady trend: not flagged (Chiefland) | **Step: flagged (I-75 & Archer Rd)**

$$F_k = \frac{\mathrm{RSS}_1 - \mathrm{RSS}_2^{(k)}}{\mathrm{RSS}_2^{(k)}/(n-3)}, \qquad F^* = \max_{k \in \{2015,\ldots,2022\}} F_k$$

$$F^* \geq F_{\mathrm{crit}} = Q_{0.999}\left(F^* \,\middle|\, z_t \overset{\mathrm{iid}}{\sim} \mathcal{N}(0,1)\right) = 36.3$$

It found new development on its own

- **28,986** lit cells tested · **1,493** flagged · ≈ 29 expected by chance
- #1 event, near I-75 and Archer Road: **28 → 91 nW**, step in 2016 (F = 176.8)
- That's **Celebration Pointe**: Bass Pro Shops opened there in November 2016
- #2: dark land west of Ocala lit up in 2021 (0.5 → 53.5 nW)
- Nobody told the model where to look; it knows light changed, not what was built

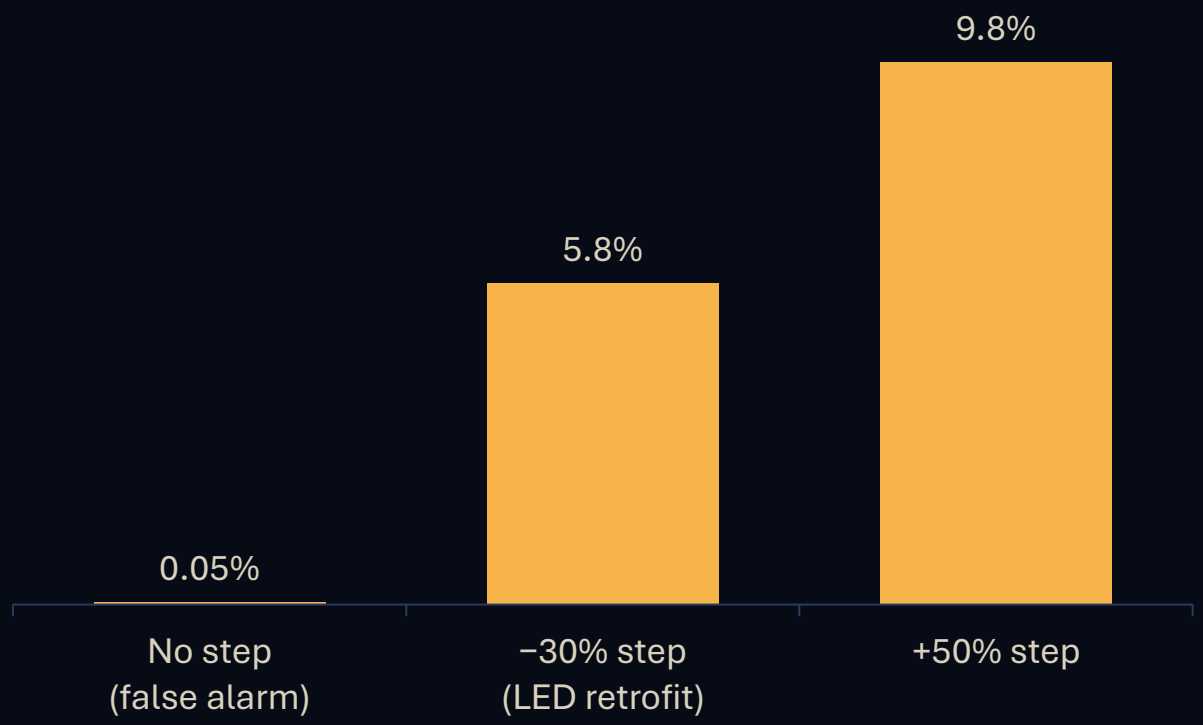


AI tab: clicking the top event flies the map to it and shows the cell's 13 years with the fitted step.

What the detector can't see

- A lit cell's yearly values scatter **≈ 10%** around its trend
- At that noise a +50% step is caught **10%** of the time, a −30% step **6%**
- So it finds large new lighting and **misses most LED retrofits** (20–40% drops)
- VIIRS is nearly blind below 500 nm: it misses most of a white LED's blue
- Next: monthly data and utility work orders

Chance a step is flagged, at the median noise (9.7%)



From 20,000 simulated series per case with a step in 2019. Large new development (+100% and more) is what it reliably finds.

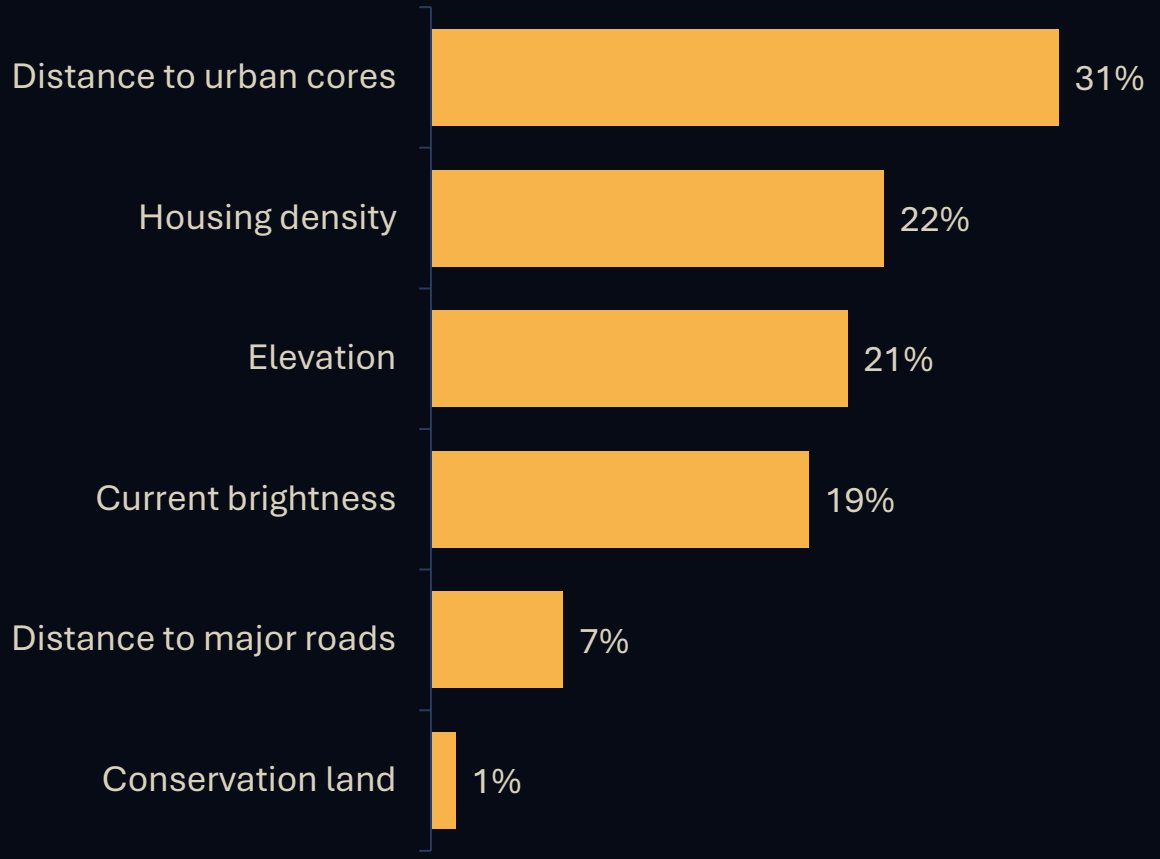
Forecasting growth with LightGBM

- Gradient-boosted decision trees on **~30,000** one-km cells
- Learns how fast a cell brightened over 6 years from **6 open inputs** describing it at the start
- Why not deep learning? Small tabular data: trains in **seconds** on a laptop, no GPU, reports what it relies on

$$F_m(\mathbf{s}) = F_{m-1}(\mathbf{s}) + \eta\, f_m(\mathbf{s}), \qquad f_m = \arg\min_f \sum_i \big(r_i - F_{m-1}(\mathbf{s}_i) - f(\mathbf{s}_i)\big)^2$$

300 trees · learning rate 0.05 · ≤ 31 leaves · ≥ 40 cells per leaf

What the model relies on (share of split gain)

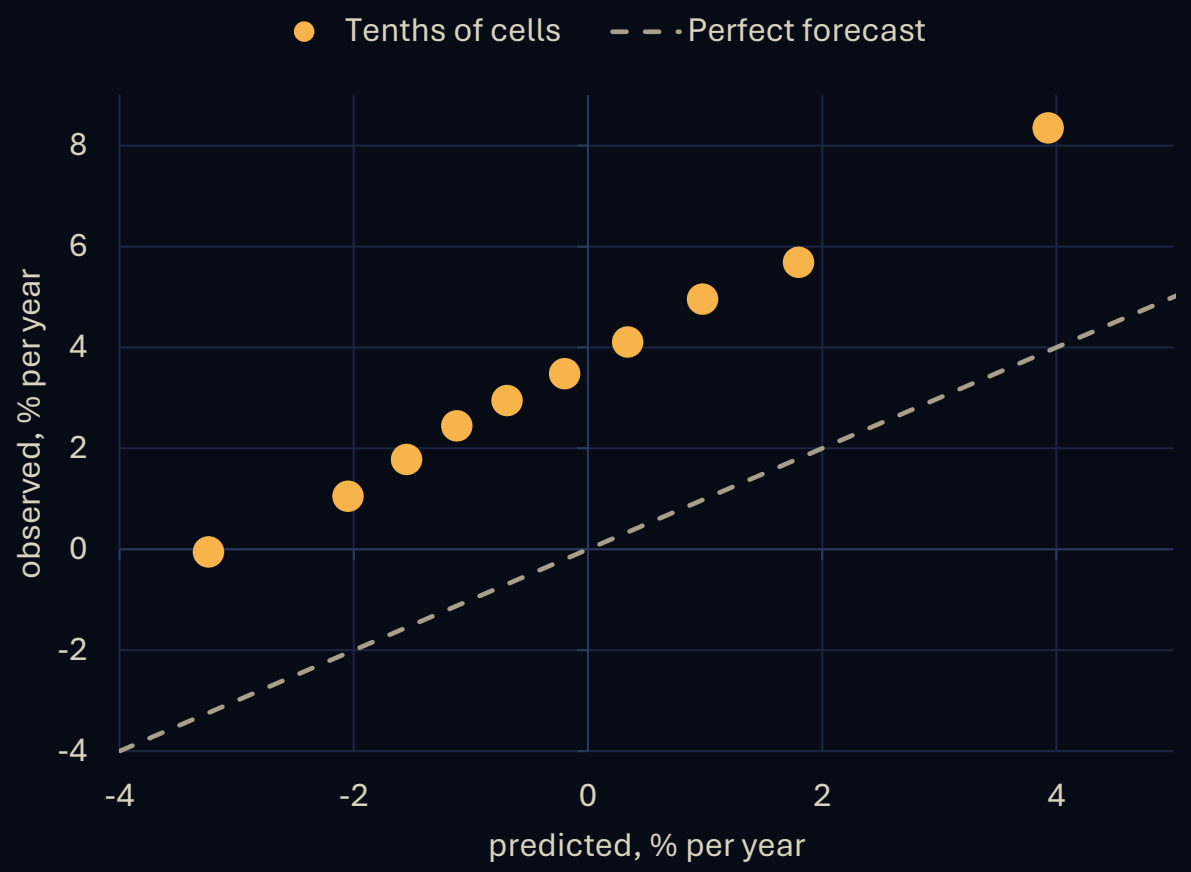


The back-test: did it beat “no change”?

- Trained on 2012 → 2018, asked for **2024**: years it never saw
- Ties “no change” on brightness: lit areas fell **3.7%** (2012–18), then rose **20.6%** (2018–24)
- But it gets the **order** right: correlation **0.35**; R^2 **0.18** on held-out areas

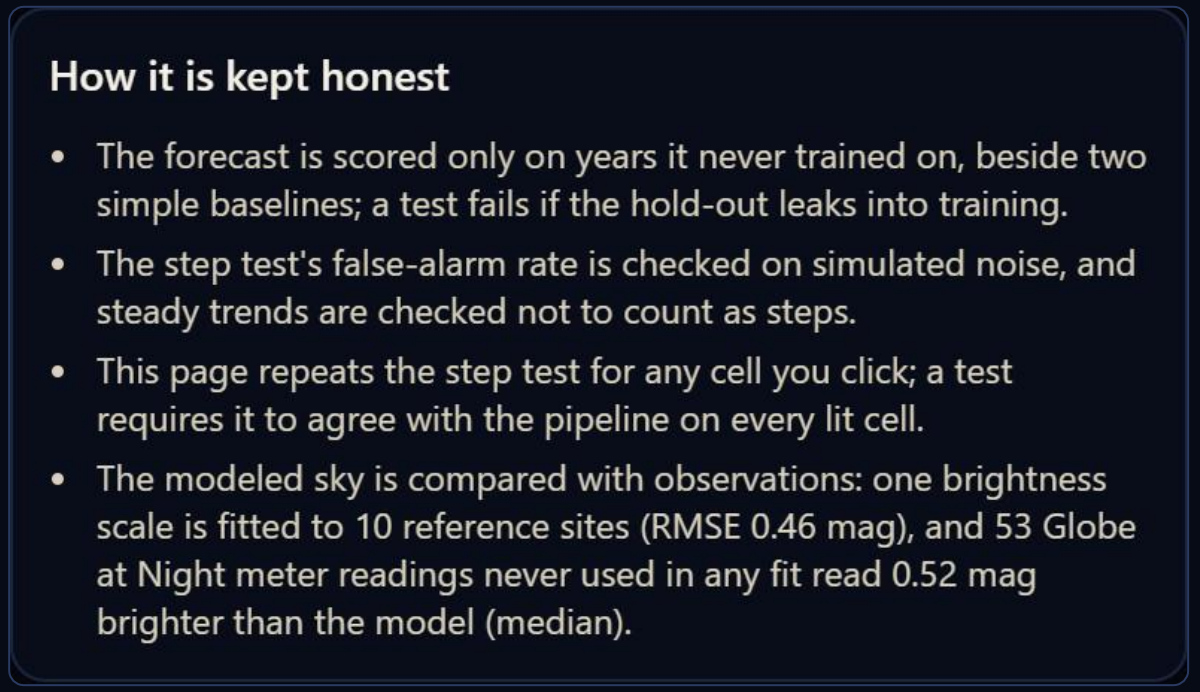
Prediction of 2024	Median error	Log error
LightGBM	**21.2%**	**0.449**
No change (2024 = 2018)	21.2%	0.467
Each cell's own past trend	25.4%	–

Predicted vs observed growth, 2018–24, by tenth of cells



How the AI is kept honest

- Forecast scored only on years it never trained on; a **unit test fails** if they leak into training
- Detector's false-alarm rate checked on simulated noise; steady trends must not count as steps
- The web page **re-runs the step test** for any cell you click; it must match the pipeline on every lit cell
- Sky model vs **53** Globe at Night meter readings never used in any fit: median 0.52 mag

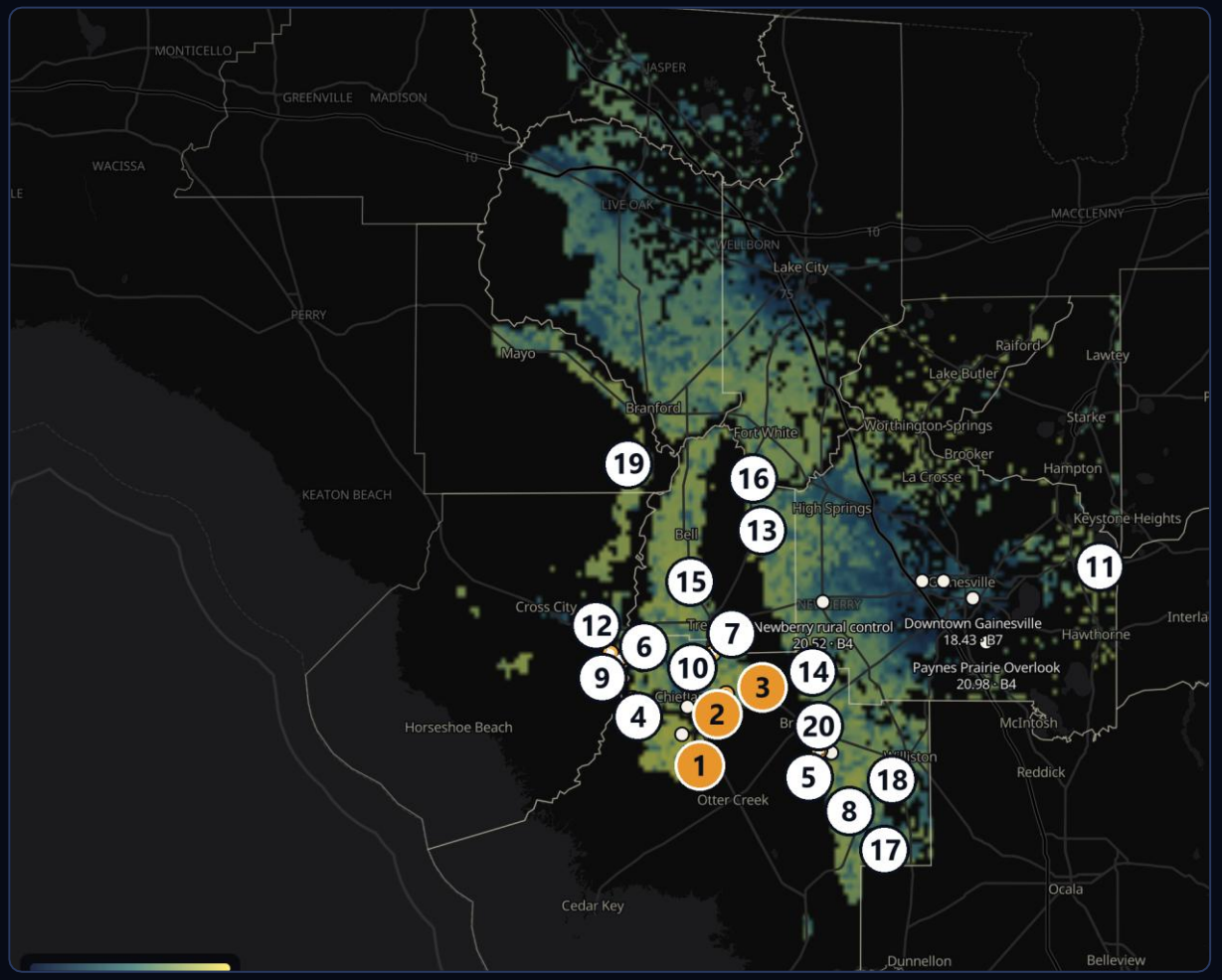


The same checks, stated in the app's AI tab.

IN USE

Where could UF build its next observatory?

- Requested by **Dr. Elizabeth Lada**, Chair of UF Astronomy
- Every ~1 km cell of 11 counties scored: dark sky today and in **2034** (LightGBM), clear air, land, access, **drive time from UF**
- Ruled out: **water and wetland**, the coast, ground below 5 m (storm surge), **2080 flood projections**, **FEMA's 100-year floodplain**, over ~90 min away
- **Shortlist: Levy #1–#3**, between Chiefland and Bronson: 44–58 min from campus, Bortle 3, land **$605K–806K** for 80 acres
- Telescope + building: $1–3M in published examples → **≈ $1.6–3.8M** total

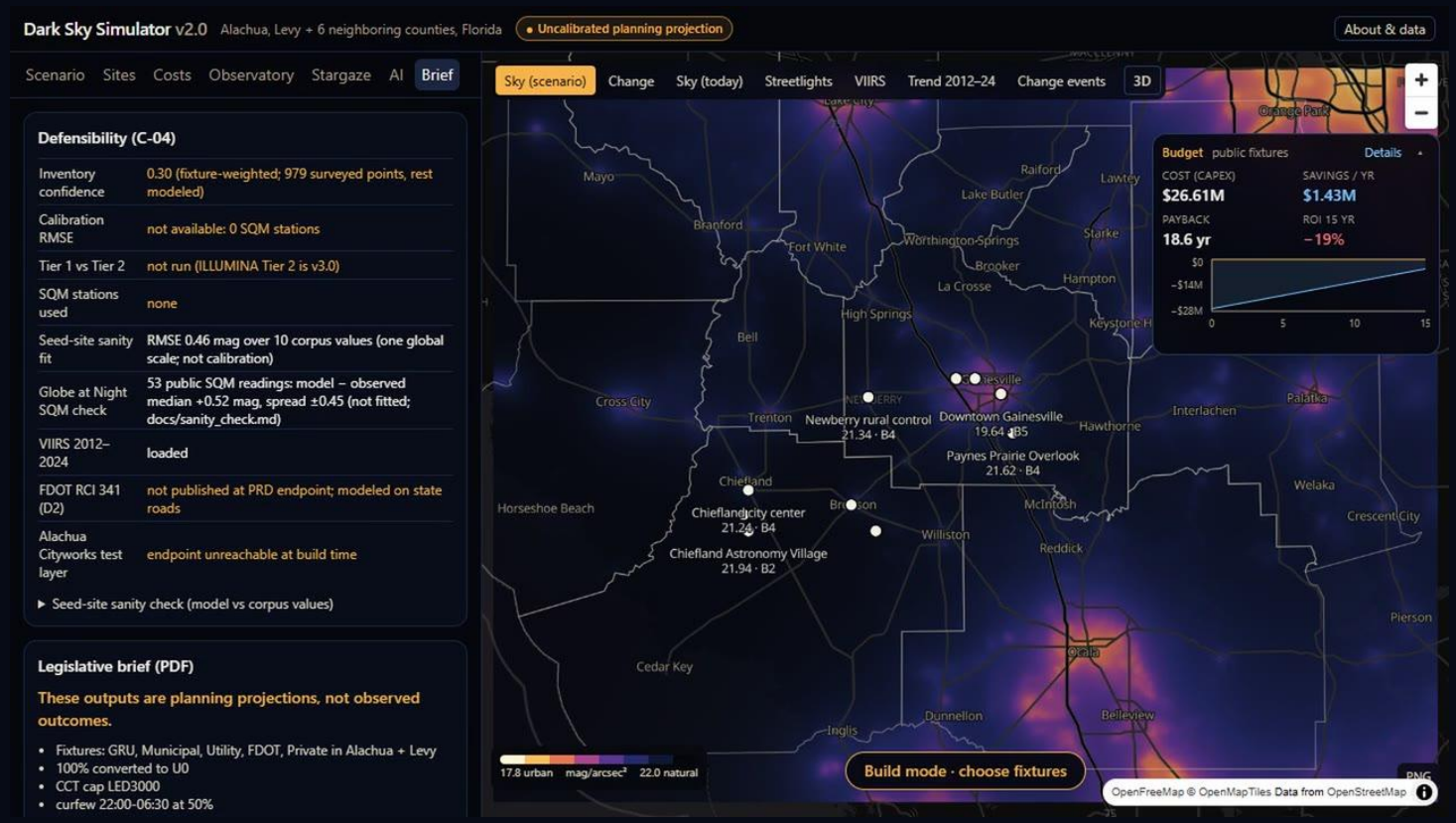


Observatory tab: the top 20 sites by rank (orange = Levy shortlist; markers enlarged). Blank areas are ruled out: water, wetland, flood-prone ground or over 90 minutes away.

IN USE

From model to policy brief

- One-click **PDF** for commissioners; every figure links back to its scenario
- First line: “These outputs are planning projections, not observed outcomes.”
- Defensibility panel lists what's missing: sky-meter stations, utility inventories, satellite calibration
- That list is the **v3 plan**

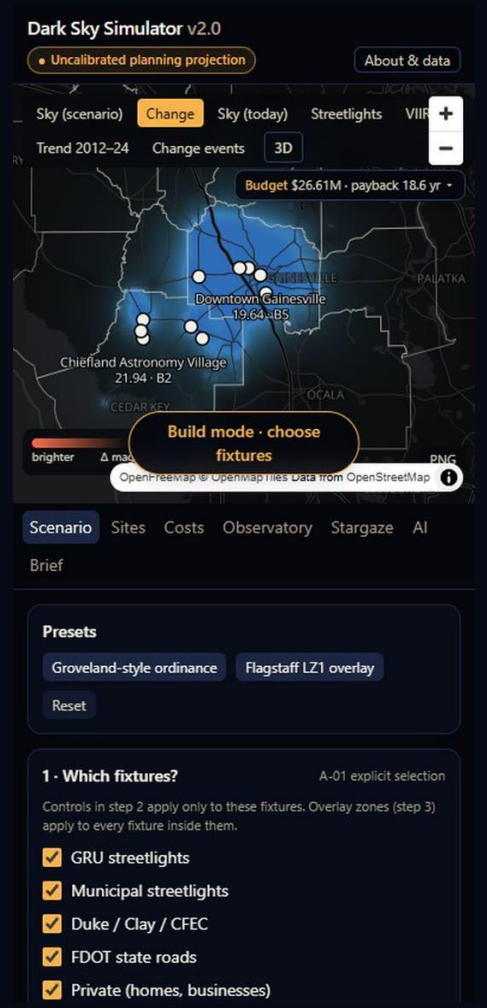


Brief tab: defensibility panel and the legislative brief.

WHAT'S NEXT

Scaling up: from projection to evidence

- **Ground truth**: sky-quality meters in every county + a 50-node low-cost sensor pilot (~$5,900 hardware)
- **Real inventories**: utility data-sharing agreements (GRU, Duke, Clay, CFEC)
- **Computer vision**: a YOLO detector on street-level imagery to classify fixtures
- **Retrofit detection**: monthly satellite data; fit the LED correction on GRU's 3000 K conversions
- **Scaling**: a reusable county package (satellite data, lighting, development, policy), Levy County next
- **Siting**: parcel-level flood maps and site visits; blind test with UF Astronomy
- Looking for collaborators in sensors, computer vision, remote sensing and policy



The same app on a phone

WHAT'S NEXT

What UF can do

- Alachua County is writing dark-sky lighting into its plan; **UF answers to the State**, so campus lighting is UF's own decision
- **Lead on campus**: upgrading UF's 4,100 pole lights costs about **$984K** and saves about **$198K** a year
- **Protect UF's skies**: keep Rosemary Hill dark and choose a site for UF Astronomy's next observatory
- **Measure the real sky**: a 50-sensor network (about **$6,000**) to check the model against real readings
- Independent and unfunded so far: I'd like this to become a UF project



UF's Rosemary Hill Observatory. Photo: The Gainesville Sun, April 2012.



Photo: Jay Rosen, May 2023

Thank you

Questions?

Try it

dark-sky-simulator-v2.vercel.app/dark-sky

Code

github.com/jayrosen-design/dark-sky-simulator-v2

Jay Rosen · jayrosen@ufl.edu

SOURCES

References

ABC7 Los Angeles (c. 2011). Domes put in place for CSUSB's observatory.

Alachua County (2026, February 13). Comprehensive Plan proposed evaluation and appraisal based amendments, Conservation and Open Space Element, Policy 5.3.7.

Andrews, D. W. K. (1993). Tests for parameter instability and structural change with unknown change point. Econometrica, 61(4), 821–856.

Bortle, J. E. (2001). Introducing the Bortle dark-sky scale. Sky & Telescope, 101(2), 126–129.

Cangialosi, J. P., & Alaka, L. (2026). Tropical Cyclone Report: Hurricane Idalia (AL102023), 26–31 August 2023. National Hurricane Center (original report February 13, 2024).

City of Groveland, Florida (2022). Ordinance 2022-26, outdoor lighting regulations.

Dark Sky Simulator fixture catalog (2026): installed unit costs from the research reports and a retail price survey of DarkSky-compliant fixture types (September 26, 2026); DarkSky Approved luminaire search, darksky.org.

Dark Sky Simulator research reports (2025): Gainesville Dark Sky Lighting Assessment; Streetlight Retrofit Cost-Benefit Analysis (Levy); Smart Streetlight Cost-Benefit Analysis; Street Lighting Dimming Savings Calculation (Duke LS-1, Clay Schedule L, CFEC). Unpublished.

DarkSky International & Illuminating Engineering Society (2020). Five Principles for Responsible Outdoor Lighting. darksky.org.

DarkSky International. DarkSky Approved program and seal. darksky.org/what-we-do/darksky-approved.

DarkSky International. International Dark Sky Places program guidelines (Dark Sky Park: sky ≥ 21.2 mag/arcsec^2). darksky.org.

Dewitz, J. (2023). National Land Cover Database (NLCD) 2021 Products. U.S. Geological Survey data release. doi:10.5066/P9JZ7AO3.

Elvidge, C. D., Zhizhin, M., Ghosh, T., Hsu, F.-C., & Taneja, J. (2021). Annual time series of global VIIRS nighttime lights derived from monthly averages: 2012 to 2019. Remote Sensing, 13(5), 922.

EMJ Construction (2016, September 1). Two new Florida Bass Pro locations underway. emjcorp.com.

Falchi, F., Cinzano, P., Duriscoe, D., et al. (2016). The new world atlas of artificial night sky brightness. Science Advances, 2(6), e1600377.

Federal Emergency Management Agency (2026). National Flood Hazard Layer (NFHL): Special Flood Hazard Areas. hazards.fema.gov, accessed October 20, 2026.

Florida Department of Revenue (2025). Tax roll and qualified sales (parcels ≥ 5 acres), via the Florida Geographic Information Office statewide parcels.

Florida Institute of Technology (2004, August 19). Funding secured for Florida Tech telescope. Florida Tech News.

Garstang, R. H. (1986). Model for artificial night-sky illumination. Publications of the Astronomical Society of the Pacific, 98, 364–375.

Gorelick, N., Hancher, M., Dixon, M., et al. (2017). Google Earth Engine: Planetary-scale geospatial analysis for everyone. Remote Sensing of Environment, 202, 18–27.

Ke, G., Meng, Q., Finley, T., et al. (2017). LightGBM: A highly efficient gradient boosting decision tree. Advances in Neural Information Processing Systems, 30, 3146–3154.

Krisciunas, K., & Schaefer, B. E. (1991). A model of the brightness of moonlight. Publications of the Astronomical Society of the Pacific, 103, 1033–1039.

Kyba, C. C. M., Altıntaş, Y. Ö., Walker, C. E., & Newhouse, M. (2023). Citizen scientists report global rapid reductions in the visibility of stars from 2011 to 2022. Science, 379(6629), 265–268.

OpenStreetMap contributors (2026). Road, street-lamp and sports-venue data. openstreetmap.org (ODbL).

Roldán Hernández, C. (2023, January 17). Security enhancements make for a brighter, safer campus. UF News, University of Florida.

Román, M. O., Wang, Z., Sun, Q., et al. (2018). NASA's Black Marble nighttime lights product suite. Remote Sensing of Environment, 210, 113–143.

Rosen, J. (2025). Dark Sky Simulator (v1) [software]. Zenodo. doi:10.5281/zenodo.17252185.

Rosen, J. (2026). Dark Sky Simulator v2 [software], incl. ADR 0013 (the 19% reconstruction). github.com/jayrosen-design/dark-sky-simulator-v2.

Rosen, J. (2026, January 13). Preserving natural skies in Alachua County: Light pollution mitigation and economic impact. Alachua County Environmental Protection Advisory Committee.

Smith, J. (2026, March 24). Restoring the Milky Way: Arts alumnus develops app to reduce light pollution. UF College of the Arts.

Sweet, W. V., et al. (2022). Global and Regional Sea Level Rise Scenarios for the United States. NOAA Technical Report NOS 01. National Oceanic and Atmospheric Administration.

The Gainesville Sun (2012, April). Photograph of the University of Florida Rosemary Hill Observatory.

U.S. Census Bureau (2020). TIGER/Line boundaries and roads; 2020 Census housing-unit counts.

U.S. Energy Information Administration (2023). How much electricity does an American home use? FAQ (2022 average: 10,791 kWh per residential customer). eia.gov.

Ward, P. J., et al. (2020). Aqueduct Floods Methodology. Technical Note. World Resources Institute. Data: WRI Aqueduct Floods Hazard Maps V2 via Google Earth Engine.

York University (2019, August 16). Installation of new telescope expected to be largest at a Canadian university. YFile.